

1 Beginnings

Throughout, F is a commutative field with $\text{char } F \neq 2$, and $a, b \in F^\times$. We follow Voight, *Quaternion Algebras*, Chapter 2.

In Mathlib the quaternion algebra is $\mathbb{H}[R, c_1, c_2, c_3]$ with relations $i^2 = c_1 + c_2i$, $j^2 = c_3$, $ij = k$ and $ji = c_2j - k$. Voight's $\left(\frac{a,b}{F}\right)$ is the special case $c_2 = 0$, i.e. $\mathbb{H}[F, a, 0, b]$, for which $i^2 = a$, $j^2 = b$, $ij = k$, $ji = -k = -ij$.

1.1 Quaternion algebras

Definition 1.1 (Quaternion algebra). Lean: `QuaternionAlgebra`

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For $a, b \in F^\times$, the *quaternion algebra* $\left(\frac{a,b}{F}\right)$ is the F -algebra $\mathbb{H}[F, a, 0, b]$ with underlying F -vector space spanned by $1, i, j, k$ subject to $i^2 = a$, $j^2 = b$, $ji = -ij$, and $k := ij$.

The relations $i^2 = a$, $j^2 = b$, $ji = -ij$ together with $k := ij$ determine the entire multiplication table by linearity and associativity: for instance $k^2 = (ij)(ij) = i(ji)j = -i^2j^2 = -ab$, and $ik = aj$, $jk = -bi$. In particular $\dim_F \left(\frac{a,b}{F}\right) = 4$, with F -basis $1, i, j, k$.

Lemma 1.2 (Standard generators presentation). Lean: `QuaternionAlgebras.standard_generators_presentation`

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An F -algebra B is a quaternion algebra if and only if there exist nonzero $i', j' \in B$ that generate B as an F -algebra and satisfy $(i')^2 = a$, $(j')^2 = b$, and $i'j' = -j'i'$ for some $a, b \in F^\times$. In that case $1, i', j', i'j'$ is automatically an F -basis of B , so $B \cong \left(\frac{a,b}{F}\right)$.

Proof. It suffices to prove that $1, i', j', i'j'$ are linearly independent. Suppose $\alpha = t + xi' + yj' + zi'j' = 0$ with $t, x, y, z \in F$. Using the relations, $0 = i'(\alpha i' + i'\alpha) = 2a(t + xi')$, and since $\text{char } F \neq 2$ and $a \neq 0$ we get $t + xi' = 0$; repeating with j' and $i'j'$ gives $t + yj' = t + zi'j' = 0$. Subtracting these from α yields $-2t = 0$, so $t = 0$ (as $B \neq 0$), whence $xi' = yj' = zi'j' = 0$; since i', j' are nonzero and $a, b \neq 0$, also $x = y = z = 0$. $\square$

1.2 Matrix representations

Every quaternion algebra can be realized as a subalgebra of 2×2 matrices over an at most quadratic extension of F .

Proposition 1.3 (Matrix embedding). Lean: `QuaternionAlgebras.matrixHom_injective`

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Let $B = \left(\frac{a,b}{F}\right)$ and let $F(\sqrt{a})$ be a splitting field over F for $x^2 - a$, with root $\sqrt{a}$. The F -linear map $\lambda: B \rightarrow M_2(F(\sqrt{a}))$ determined by

$$i \mapsto \begin{pmatrix} \sqrt{a} & 0 \\ 0 & -\sqrt{a} \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix}, \quad t + xi + yj + zk \mapsto \begin{pmatrix} t + x\sqrt{a} & b(y + z\sqrt{a}) \\ y - z\sqrt{a} & t - x\sqrt{a} \end{pmatrix}$$

is an injective F -algebra homomorphism and an isomorphism onto its image.

Proof. The matrices $\lambda(i), \lambda(j)$ satisfy $\lambda(i)^2 = a$, $\lambda(j)^2 = b$ and $\lambda(j)\lambda(i) = -\lambda(i)\lambda(j)$, so by the defining relations λ is an F -algebra homomorphism. For injectivity, $\lambda(t + xi + yj + zk) = 0$ forces every matrix entry to vanish, and using the linear independence of $1, \sqrt{a}$ over F this gives $t = x = y = z = 0$. $\square$

Corollary 1.4 (Split quaternion algebra is a matrix algebra). `Lean: QuaternionAlgebras.split_matrix`

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There is an F -algebra isomorphism $\left(\frac{1,b}{F}\right) \xrightarrow{\sim} M_2(F)$, given by $i \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $j \mapsto \begin{pmatrix} 0 & b \\ 1 & 0 \end{pmatrix}$. In particular $\left(\frac{1,1}{F}\right) \cong M_2(F)$.

Proof. Specialize 1.3 with $a = 1$, so $\sqrt{a} = 1 \in F$ and λ lands in $M_2(F)$. It is an injective F -algebra homomorphism between F -spaces of equal dimension $4 = \dim_F M_2(F)$, hence surjective, hence an F -algebra isomorphism. $\square$

Lemma 1.5 (Left regular representation and its faithfulness). `Lean: QuaternionAlgebras.leftRegRepK,`

`QuaternionAlgebras.leftRegRepK_faithful`

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Let $K = F[i] \cong F[x]/(x^2 - a)$, and regard $B = \left(\frac{a,b}{F}\right)$ as a right K -module via $\beta \cdot k = \beta \iota(k)$ (well defined since K is commutative). The *left regular representation* of B (Voight (2.3.9)) is the ring homomorphism $\lambda: B \rightarrow \text{End}_K(B)$, $\alpha \mapsto \lambda_\alpha$, where $\lambda_\alpha(\beta) = \alpha\beta$; each λ_α is K -linear because left and right multiplication commute. This representation is injective (a *faithful* representation).

Proof. If $\lambda_\alpha = 0$ then $\alpha = \alpha \cdot 1 = \lambda_\alpha(1) = 0$. $\square$

Lemma 1.6 (The Hamiltonian embedding into $M_2(\mathbb{C})$). `Lean: QuaternionAlgebras.hamiltonToComplex,`

`QuaternionAlgebras.hamiltonToComplex_injective, QuaternionAlgebras.hamiltonToComplex_trace, QuaternionAlgebras.hamiltonToComplex_normSq`

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The real Hamiltonians $\mathbb{H} = \left(\frac{-1,-1}{\mathbb{R}}\right)$ embed into $M_2(\mathbb{C})$ (Voight (2.4.1)) via

$$i \mapsto \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad t + xi + yj + zk \mapsto \begin{pmatrix} t + xi & -y - zi \\ y - zi & t - xi \end{pmatrix},$$

the matrix embedding with $a = b = -1$ and $\sqrt{-1} = i \in \mathbb{C}$. This embedding λ is injective, and (Voight (2.4.7)) for $\alpha = t + xi + yj + zk$,

$$\text{tr}(\lambda(\alpha)) = 2t = 2 \text{Re}(\alpha), \quad \det(\lambda(\alpha)) = t^2 + x^2 + y^2 + z^2 = \|\alpha\|^2.$$

Proof. Injectivity is the specialization of the matrix embedding (1.3) to $a = b = -1$ over $\mathbb{C} = \mathbb{R}(\sqrt{-1})$. For the trace and determinant, the displayed matrix has trace $(t + xi) + (t - xi) = 2t$ and determinant $(t + xi)(t - xi) - (-y - zi)(y - zi) = (t^2 + x^2) + (y^2 + z^2)$, the reduced norm $\|\alpha\|^2 = \text{normSq}(\alpha)$. $\square$

1.3 Rotations

We work with the real Hamiltonians $\mathbb{H} = \left(\frac{-1, -1}{\mathbb{R}}\right)$, the unit group $\mathbb{H}^1 = \{\alpha : \|\alpha\|^2 = 1\}$, and $\mathbb{H}^0 \cong \mathbb{R}^3$ the pure quaternions. Following Voight 2.4, we set up conjugation, the norm, and the pure quaternions, and then use unit quaternions to model rotations of $\mathbb{R}^3$.

The *conjugation* (standard involution) on $\left(\frac{a, b}{F}\right)$ is Mathlib's `star` operation, $\alpha = t + xi + yj + zk \mapsto \bar{\alpha} = t - xi - yj - zk$. One computes $\alpha + \bar{\alpha} = 2t$ and the norm $\alpha\bar{\alpha} = \bar{\alpha}\alpha = t^2 + x^2 + y^2 + z^2 =: \|\alpha\|^2$, which agrees with the Euclidean square norm on $\mathbb{H} \cong \mathbb{R}^4$ and is multiplicative: $\|\alpha\beta\|^2 = \|\alpha\|^2\|\beta\|^2$.

A quaternion α is *pure* (imaginary) if its real part vanishes, equivalently $\bar{\alpha} = -\alpha$, i.e. α is *skew-adjoint* for the standard involution. We do not introduce a bespoke definition: the pure quaternions are Mathlib's skew-adjoint submodule `skewAdjoint.submodule` $\mathbb{R}\mathbb{H}$, $\mathbb{H}^0 = \text{skewAdjoint}(\mathbb{H}) = \mathbb{R}i + \mathbb{R}j + \mathbb{R}k \cong \mathbb{R}^3$; dually the *real* quaternions are the self-adjoint submodule `selfAdjoint.submodule` $\mathbb{R}\mathbb{H}$, $\text{selfAdjoint}(\mathbb{H}) = \mathbb{R} \cong \mathbb{R}$ (those with $\bar{\alpha} = \alpha$).

Lemma 1.7 (Product of pure quaternions). Lean: `QuaternionAlgebras.pure_product`

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For pure quaternions $v = v_1i + v_2j + v_3k$ and $w = w_1i + w_2j + w_3k$ in $\mathbb{H}^0 \cong \mathbb{R}^3$,

$$vw = -(v \cdot w) + (v \times w),$$

where $v \cdot w$ is the dot product and $v \times w$ the cross product (a pure quaternion).

Proof. Expand vw over the basis i, j, k using $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$. The diagonal terms contribute the real part $-(v_1w_1 + v_2w_2 + v_3w_3) = -(v \cdot w)$, and the off-diagonal terms collect to $(v_2w_3 - v_3w_2)i + (v_3w_1 - v_1w_3)j + (v_1w_2 - v_2w_1)k = v \times w$. $\square$

Lemma 1.8 (Pure quaternions: square, orthogonality, and anticommutation). Lean: `QuaternionAlgebras.pure_square`, `QuaternionAlgebras.pure_mul_isSkewAdjoint_iff`, `QuaternionAlgebras.pure_anticommut_iff`

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Let v, w be pure quaternions. (1) A quaternion v is pure if and only if $v^2 = -\|v\|^2$; in particular for pure v , $v^2 = -\|v\|^2 \in \mathbb{R}_{\leq 0}$. (2) The product vw is pure if and only if v, w are orthogonal. (3) One has $wv = -vw$ if and only if v, w are orthogonal.

Proof. Part (1) is Mathlib's `Quaternion.sq_eq_neg_normSq`: $v^2 = -\text{normSq } v \iff v.\text{re} = 0$. For (2) and (3), by 1.7 $vw = -(v \cdot w) + v \times w$ and $wv = -(v \cdot w) - v \times w$ (the cross product is antisymmetric). Hence the real part of vw is $-(v \cdot w)$, which vanishes iff $v \cdot w = 0$, giving (2); and $wv + vw = -2(v \cdot w)$, so $wv = -vw$ iff $v \cdot w = 0$, giving (3). $\square$

Definition 1.9 (Rotation angle). Lean: `QuaternionAlgebras.rotAngle`

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The *rotation angle* of a unit quaternion $\alpha \in \mathbb{H}^1$ is $\theta := \arccos(\text{Re } \alpha) \in [0, \pi]$ (Voight (2.4.16)), so that $\alpha = \cos \theta + (\sin \theta)I(\alpha)$.

Definition 1.10 (Axis of a unit quaternion). Lean: QuaternionAlgebras.axis

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The *axis* $I(\alpha)$ of a quaternion (Voight (2.4.16)) is the normalized pure part

$$I(\alpha) := \frac{xi + yj + zk}{\|xi + yj + zk\|}, \quad \alpha = t + xi + yj + zk.$$

For $\alpha \in \mathbb{H}^1 \setminus \{\pm 1\}$ there is a unique $\theta \in (0, \pi)$ with $\cos \theta = t$ and $\alpha = \cos \theta + (\sin \theta) I(\alpha)$; then $I(\alpha)$ is pure with $\|I(\alpha)\| = 1$ and $I(\alpha)^2 = -1$, the axis about which ρ_α rotates.

Definition 1.11 (Adjoint representation). Lean: QuaternionAlgebras.adjoint

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The *adjoint representation* (Voight (2.4.13)) is the group homomorphism

$$\text{adjoint}: \mathbb{H}^1 \rightarrow \text{Aut}(\mathbb{H}^0), \quad \alpha \mapsto \rho_\alpha, \quad \rho_\alpha: v \mapsto \alpha v \alpha^{-1},$$

sending a unit quaternion α to conjugation, an $\mathbb{R}$ -linear automorphism ρ_α of the pure quaternions $\mathbb{H}^0$ (with inverse $\rho_{\alpha^{-1}}$). It is a homomorphism since $\rho_{\alpha\beta} = \rho_\alpha \circ \rho_\beta$ and $\rho_1 = \text{id}$; that its image lands in $\text{SO}(3)$ is 1.12 and 1.15.

Proposition 1.12 (Conjugation acts by rotations). Lean: QuaternionAlgebras.rotation_mem_so, QuaternionAlgebras.ro

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For $\alpha \in \mathbb{H}^1$, the conjugation map $\rho_\alpha: \mathbb{H}^0 \rightarrow \mathbb{H}^0$, $v \mapsto \alpha v \alpha^{-1}$, is a well-defined $\mathbb{R}$ -linear map lying in $\text{SO}(3)$. Setting $\theta := \arccos(\text{Re } \alpha) \in [0, \pi]$ (Voight (2.4.16)), so $\alpha = \cos \theta + (\sin \theta) I(\alpha)$, the map ρ_α is the rotation by the angle 2θ about the axis $I(\alpha)$; concretely its $\text{SO}(3)$ matrix has trace

$$\text{tr}(\rho_\alpha) = 1 + 2 \cos 2\theta = 4(\text{Re } \alpha)^2 - 1,$$

which determines the rotation angle 2θ .

Proof. Conjugation is $\mathbb{R}$ -linear and preserves $\mathbb{H}^0$ (it preserves the real part) and the norm, since $\|\alpha v \alpha^{-1}\|^2 = \|v\|^2$ for $\alpha \in \mathbb{H}^1$; thus its matrix in the orthonormal basis i, j, k is orthogonal, and a direct computation gives $\det \rho_\alpha = 1$, so $\rho_\alpha \in \text{SO}(3)$. In that basis the diagonal of the rotation matrix sums to $3t^2 - x^2 - y^2 - z^2 = 4t^2 - 1$ using $t^2 + x^2 + y^2 + z^2 = 1$, while $1 + 2 \cos 2\theta = 1 + 2(2t^2 - 1) = 4t^2 - 1$ since $\cos \theta = t = \text{Re } \alpha$; hence ρ_α rotates by 2θ about $I(\alpha)$. $\square$

Lemma 1.13 (Surjectivity of the rotation map). Lean: QuaternionAlgebras.rotation_surjective

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Every $A \in \text{SO}(3)$ is realized by conjugation: there is a unit quaternion α , $\|\alpha\|^2 = 1$, whose rotation matrix equals A . Hence the rotation map $\mathbb{H}^1 \rightarrow \text{SO}(3)$ is surjective.

Proof. Following Voight's Exercise 2.15, A fixes a unit axis u (an eigenvector for eigenvalue 1, since $\det(A - I) = 0$ in odd dimension). Choose a unit $w \perp u$ and write $Aw = a w + b(u \times w)$ with $a^2 + b^2 = 1$; take θ with $\cos \theta = a$ and $\sin \theta = b$, and set $\alpha = \cos(\theta/2) + \sin(\theta/2) u$. Then ρ_α agrees with A on the orthonormal frame $\{u, w, u \times w\}$ — on u both fix it, on w by the choice of θ , and on $u \times w$ because both A and ρ_α preserve cross products — so $\rho_\alpha = A$. $\square$

Lemma 1.14 (Kernel of the rotation map). Lean: QuaternionAlgebras.rotation_kernel

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If a unit quaternion $\alpha \in \mathbb{H}^1$ fixes every pure quaternion under conjugation (i.e. $\rho_\alpha = \text{id}$ on $\mathbb{H}^0$), then $\alpha = \pm 1$.

Proof. If α fixes every pure quaternion under conjugation then α commutes with every pure quaternion, hence with all of $\mathbb{H}$; so α is central and therefore real. A real quaternion with $\|\alpha\|^2 = \alpha^2 = 1$ satisfies $\alpha = \pm 1$. $\square$

Corollary 1.15 (The double cover $\mathbb{H}^1 \rightarrow \text{SO}(3)$). Lean: QuaternionAlgebras.rotationHom, QuaternionAlgebras.rotationHom_ker

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The rotation map is a group homomorphism $\rho: \mathbb{H}^1 \rightarrow \text{SO}(3)$, $\alpha \mapsto \rho_\alpha$, which is surjective and has kernel $\ker \rho = \{\pm 1\}$, giving the exact sequence

$$1 \rightarrow \{\pm 1\} \rightarrow \mathbb{H}^1 \rightarrow \text{SO}(3) \rightarrow 1.$$

Proof. The assignment is a homomorphism since $\rho_{\alpha\beta} = \rho_\alpha \circ \rho_\beta$, and it lands in $\text{SO}(3)$ by 1.12. It is surjective because every element of $\text{SO}(3)$ is a rotation about some axis, realized by $\alpha = \cos \theta + (\sin \theta)u$. If α is in the kernel it commutes with every pure quaternion, hence is central, hence real with $\|\alpha\|^2 = \alpha^2 = 1$, i.e. $\alpha = \pm 1$. $\square$